1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

TLS-rich MIBC tumors fail to translate TLS architecture into effective anti-tumor immunity when tumor MSH6 deficiency reshapes neoantigen quality/quantity and fosters a PD-L1–dominant, S100A9+ myeloid-inflamed microenvironment that functionally suppresses TLS-driven T-cell activity. Specifically, MSH6-deficient, PD-L1–high tumors will exhibit: (a) impaired TLS T-cell activation and reduced TCR clonal expansion despite abundant TLS; (b) higher PD-L1 expression on tumor and/or myeloid cells and enriched S100A9+ myeloid infiltrates in TLS-adjacent zones; (c) diminished cytotoxic CD8+ T-cell effector responses and increased exhaustion markers within TLS neighborhoods; (d) lower objective/durable responses to PD-1/PD-L1 blockade compared with TLS-rich, MSH6-proficient tumors. Contextual modifiers (HDL-C, TP53 status, ER signaling, PTGDS) will modulate the strength of this suppression by affecting PD-L1 regulation, myeloid polarization, or TLS maintenance. Consequently, TLS abundance will be a poor predictor of response to PD-1/PD-L1 inhibitors specifically in the subset of MIBC tumors that are MSH6-deficient and PD-L1–high; TLS will retain predictive value in MSH6-proficient or PD-L1-low contexts. The model also predicts that strategic disruption of the myeloid/S100A9 axis or co-targeting PD-L1 will restore TLS functionality and improve therapeutic responses in preclinical systems and some patient subsets.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	MSH6, PD-L1
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSH6: MSH6/MSI) and systemic/neoantigen burden (bTMB) to local immune suppression via PD-L1 and inflammatory myeloid signals, which plausibly interfere with TLS function.

PD-L1: PD-L1 inhibitor response) is the most specific, actionable endpoint to test the mechanistic hypothesis and to guide treatment selection.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	S100A9	Supporting modifier from knowledge graph

Secondary	bTMB	Supporting modifier from knowledge graph
Secondary	MSI/dMMR	Supporting modifier from knowledge graph
Secondary	HDL-C	Supporting modifier from knowledge graph
Secondary	TP53	Supporting modifier from knowledge graph
Secondary	PTGDS	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: among TLS-rich tumors, MSH6def/PD-L1high cases show poorer PD-1/PD-L1 inhibitor responses than MSH6proficient cases;

Prediction 2: single-cell/spatial analyses reveal proximate S100A9+ myeloid cells suppressing TLS-associated T-cell activity in MSH6def/PD-L1high tumors;

Prediction 3: combining PD-L1 blockade with myeloid-targeting or metabolic modulators can rescue TLS function and improve responses;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Gene expression does not equal protein activity; phospho-protein or functional assays are needed for confirmation.

Alternative Directions

- S100A9: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- bTMB: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.